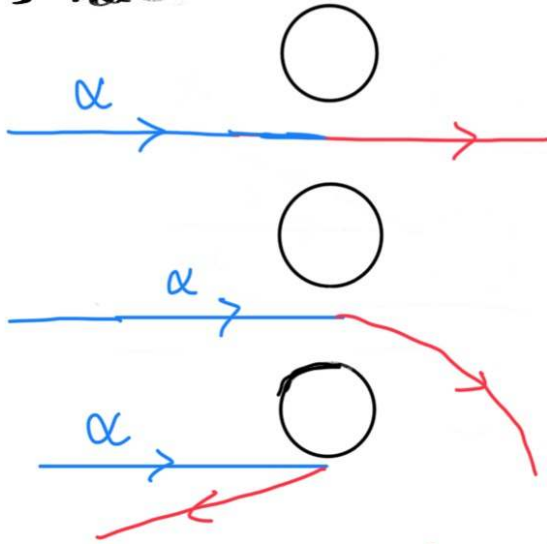


Nuclear Physics

Rutherford's golden foil theory.

* Most of α particles pass through the golden foil as most of atom's volume is empty

* Few α particles are deflected by the repulsion and very few are reflected, so there is a dense positive center (nucleus) that contains most of the atom's mass



$A \rightarrow$ nucleon number
 $n+p$

* Charge of proton: charge of electron:
 $1.6 \times 10^{-19} \text{ C}$

* atomic mass: Unit: (u) = mass of one nucleon

Z

$\hookrightarrow$ Proton number = electron number in a neutral atom

Isotopes: Elements of same proton number but differ in neutron number (nucleon number)

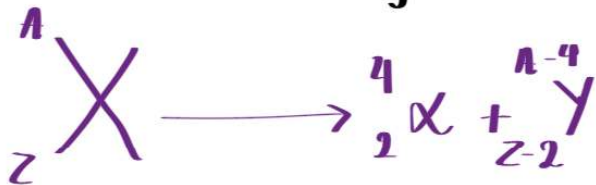
→ All isotopes have the same chemical properties because they have the same proton number.

→ They differ in physical properties, like mass & stability.

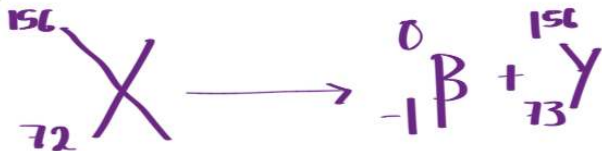
* Unstable nuclei have excess of neutrons, or are heavy nuclei, and they release radiations: alpha (α), Beta (β), Gamma (γ), to try to be more stable.

Alpha particles (${}^4_2\alpha$): They are two protons and two neutrons cracked from the unstable nuclei to be more stable. mass: 4u

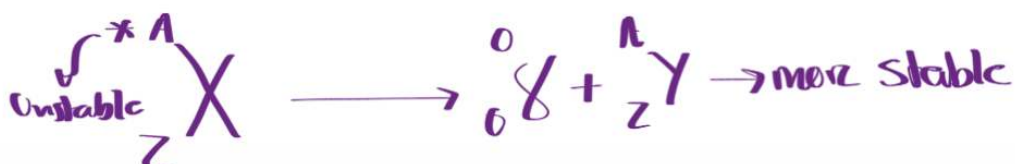
Charge: $+2e^-$



Beta particles (${}^0_{-1}\beta$): Because of the instability, one of the neutrons converts to a proton and releases an electron.



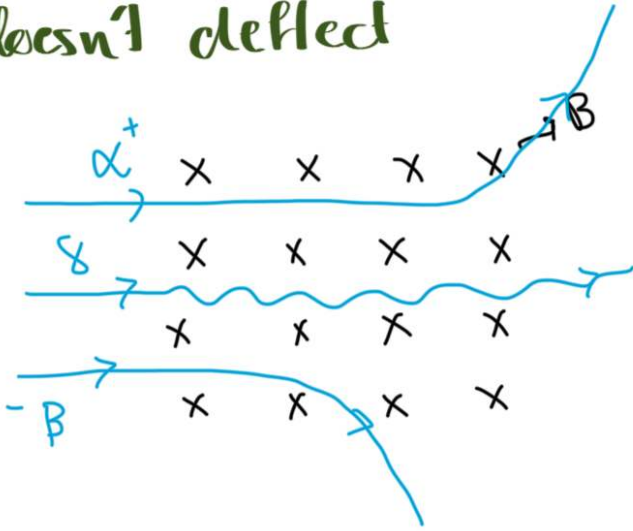
Gamma rays (${}^0_0\gamma$): An electromagnetic wave that is emitted from the nucleus.



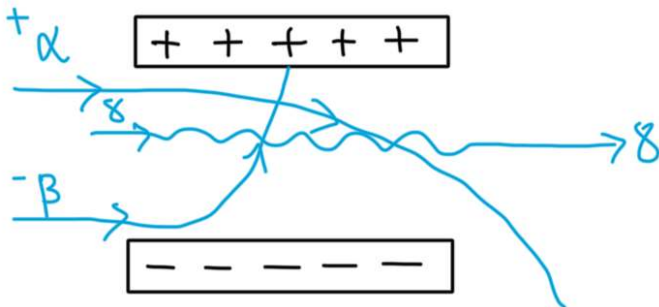
Properties of α , β , γ :

1- Deflection by magnetic field:

- α deflects weakly because it is heavy
- β deflects strongly because it is lighter
- γ doesn't deflect

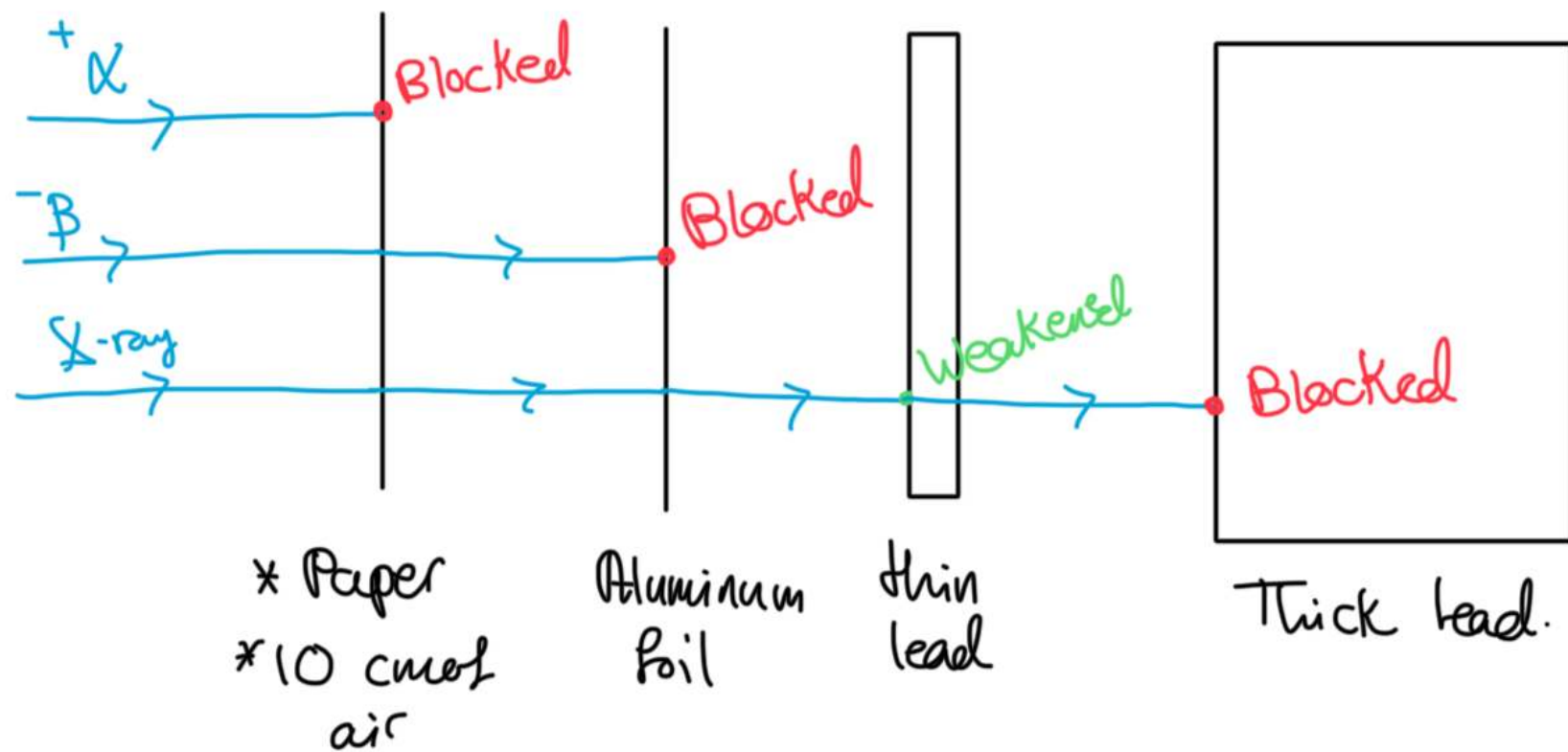


2- Deflection by electric field

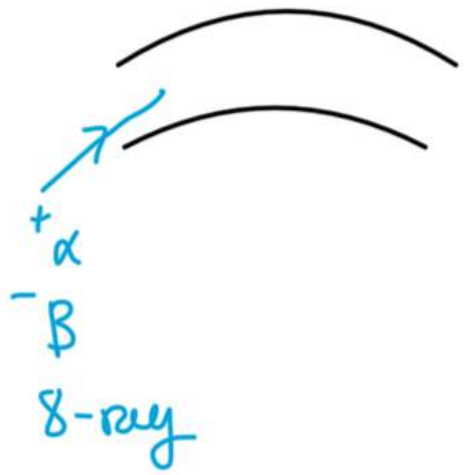


3- Penetration:

- α can be blocked by paper or 10 cm of air. So α is the worst penetrator, because it is heavy and highly charged
- β can be blocked by aluminum foil
- γ -ray can be weakened by thin lead, and blocked by thick lead. So, γ -ray is the best penetrator, because it is chargeless and massless



4-Ionization: It's the Kick out of the atom's electrons by the collision with α , β , γ



- α is the best ionizer because it is heavy and highly charged.

- γ -ray is the worst ionizer because it is chargeless and massless.

5-Hazard:

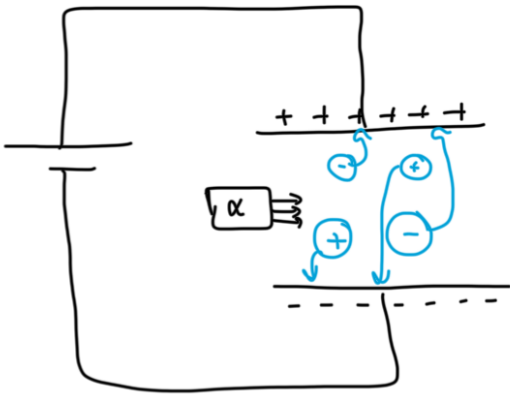
* If the source is outside the body, γ -ray is the most dangerous since it has high ability to penetrate.

* If the source is inside the body, α is the most dangerous since it has high ability to ionize.

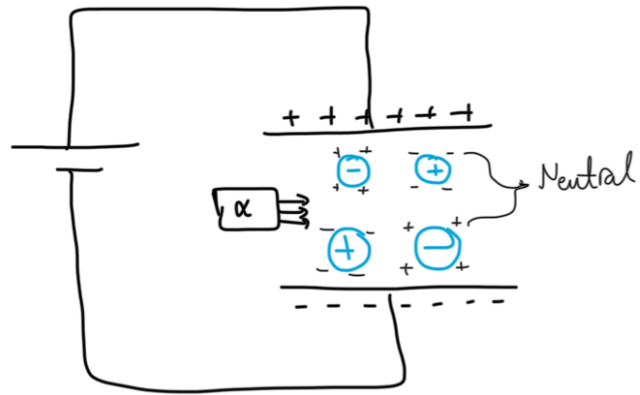
→ These ionizing radiation can cause cancer, mutation, and death

Uses of ionizing radiation:

* For α particles: Smoke alarms.



* α particles will ionize the air between the plates
→ so it is a closed circuit
→ Normal: No smoke



* When there is smoke, the smoke particles will neutralize the air → becomes an open circuit
→ there is smoke → Alarm

* For β particles: Measuring small thicknesses:

→ We are depending on the number of β that penetrated through the paper to know it's thickness.

* If the number of survivals is higher than needed → the thickness is smaller than needed.

* If the number of survivals is lower than needed → the thickness is higher than needed

* for γ -rays: - Treating cancer

- Sterilizing tools and food

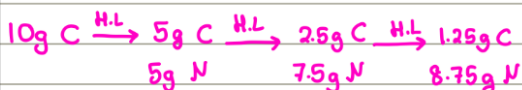
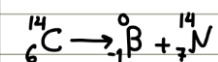
- Detecting cracks: By injecting a radioactive tracer, we can detect the radiation above the ground to locate the crack

Background radiation: It is the radiation that is coming from the surrounding radon in air/soil/space/walls.

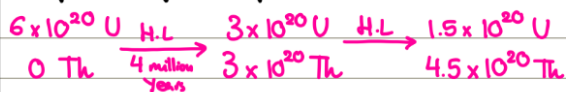
* We use a detector to detect the count rate (activity) of the elements. (count/s) (count/min)

* Background radiation is almost constant since it is a random process.

Half-life: The time needed for half of the nuclei/sample to decay.



Example: In a pure sample of ${}^{238}\text{U}$ (6×10^{20} nuclei). Uranium decays to form thorium with half-life = 4 million years



After 8 million years: 2 Half lives:

Remaining: $1.5 \times 10^{20} \text{ U}$

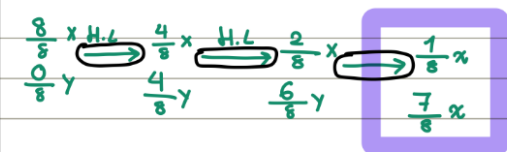
The produced = $4.5 \times 10^{20} \text{ Th}$

Number of emitted α -particles: 4.5×10^{20}

Example: If (X) decays to (Y) with H.L = 2 weeks. In a pure sample, it was found that 7 nuclei of (Y) for each of (X). Find the age of the sample.

X: Y

1: 7 $\rightarrow$ 8



3 steps $\rightarrow 3 \times 2 = 6$ weeks

Background radiation

Example: If the reading of a detector with NO radioactive sample is 30 count/s, and after placing a sample in front of the detector, the reading is 310 count/s. Find the reading after a year, knowing that the H.L = 4 months.

① $310 - 30 = 280$ count/s

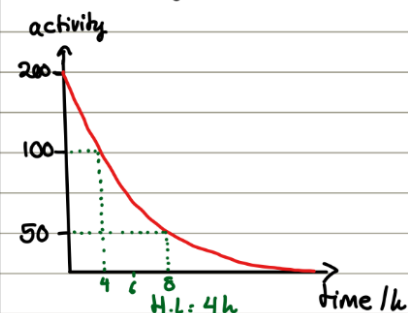
② $\frac{12}{4} = 3$ steps

③ $280 \rightarrow 140 \rightarrow 70 \rightarrow 35$

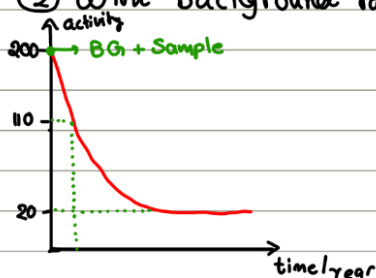
④ $35 + 30 = 65$ count/s

Decay curve

① No Background radiation:



② With background radiation



BG: 20

$200 - 20 = 180 \rightarrow$ sample only

$180 \div 2 = 90 \rightarrow$ sample only

$90 + 20 = 110 \rightarrow$ sample + BG